HELIXLINE

EVIDENCE-BASED HEALTH INTELLIGENCE

SMART

HEALTH REPORT

PERSONALIZED | PRECISE | ACTIONABLE

PREPARED FOR

Demo User

13 February 2026

Your Health Vitals

Demo User
Booking ID: 16897794339 | Sample Collection Date: 13 February 2026

Dear Demo User,

Congratulations on taking this important step towards better health. Your Helixline report is powered by a **deterministic AI health engine**, a clinically validated system combining **machine learning models**, evidence-based scoring algorithms, and age & gender adjusted reference standards. Every insight is derived *exclusively* from your actual lab values: **100% personalised, nothing assumed, nothing generalised.** Every recommendation is tailored to you. When you retest, you will see a **direct before-and-after comparison** to measure exactly how far you have come.

We are in your corner every step of the way. Team Helixline

81 HEALTH MARKERS	9 TEST PANELS	8 ORGAN SYSTEMS	Your test spans the **full spectrum of preventive health**: haematology, organ biochemistry, lipid metabolism, hormonal status, micronutrients, thyroid axis, and inflammatory biomarkers. All in a single blood draw. One of the **most comprehensive health assessments** available outside a hospital setting.

TESTS COVERED IN THIS REPORT

Hemogram / CBC 25 markers	**Lipid & Cardiac** 12 markers	**Liver Function** 12 markers
Kidney Function 12 markers	**Hormones** 8 markers	**Metabolic & Diabetes** 4 markers
Thyroid Panel 3 markers	**Vitamins & Bone** 3 markers	**Inflammation Markers** 2 markers

10 Vital Health Parameters of the Human Body Ecosystem

Core indicators for routine primary healthcare, with personalised values from your test results.

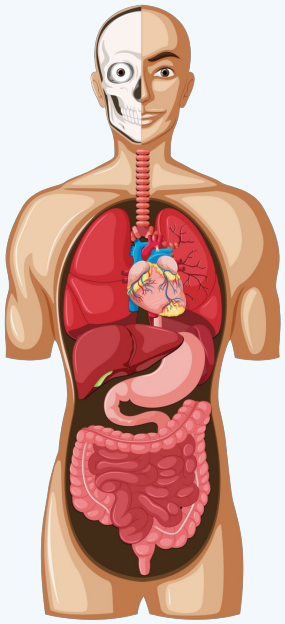
Hba1c (Glycosylated Hemoglobin)
6.2 %
• Subpar

Thyroid Stimulating Hormone (TSH)-Ultrasensit
3.227 µIU/ml
• Optimal

GFR, ESTIMATED
110.6 mL/min/1.73m2
• Optimal

Haemoglobin (HB)
13.7 g/dL
• Optimal

Total Cholesterol
177 mg/dl
• Optimal



VITAMIN D (25 - OH VITAMIN D)
26.69 ng/ml
• Subpar

Serum iron
65 ug/dl
• Subpar

VITAMIN B12
491 pg/ml
• Optimal

Serum Albumin
4.31 g/dl
• Optimal

Serum Calcium
9.5 mg/dl
• Optimal

Disclaimer
This report provides interpretative health insights based on laboratory data and is not a medical diagnosis, treatment plan, or prescription. Clinical decisions should be made in consultation with a qualified healthcare provider.

Predictive Health Indexes

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These are scientifically validated health scores calculated from your blood results. Unlike looking at a single test value, each index combines several markers to reveal hidden health risks that a routine report may miss. They are powerful early-warning tools used by doctors worldwide to guide preventive health decisions.

CARDIOVASCULAR

D'Agostino et al. JAMA 2008

Framingham 10-Yr CVD Risk

10-Year Cardiovascular Event Probability

IN PLAIN TERMS Your personal probability of having a heart attack or stroke in the next 10 years. It combines age, cholesterol, blood pressure, and lifestyle factors into one number to guide preventive treatment decisions.

5.4
Scored out of 30%

Low 10-year CVD risk (<10%) (assumed: blood pressure not measured)

Estimates your probability of a heart attack, stroke, or cardiovascular death in the next 10 years. Accounts for age, lipid levels, blood pressure, smoking status, and diabetes to give a precise, individualised risk figure.

Derived from the Framingham Heart Study. Accounts for age, total cholesterol, HDL, systolic BP, smoking, and diabetes. Guides primary prevention therapy decisions worldwide.

CARDIOVASCULAR

Dobiasova & Frohlich Clin Biochem 2004

AIP

Atherogenic Index of Plasma

IN PLAIN TERMS This ratio captures the balance between artery-clogging fat particles and protective cholesterol. An unfavourable balance can drive plaque formation and heart attacks even when your LDL appears normal on a standard panel.

0.159
Scored out of 0.5

Intermediate atherogenic risk

Reflects the balance between fat particles that can clog arteries and protective HDL cholesterol. Helps identify dangerous small, dense cholesterol particles that standard tests cannot directly measure.

Formula: $\log_{10}$(TG mmol/L ÷ HDL mmol/L). A powerful predictor of coronary artery disease independent of total cholesterol and LDL. Requires only TG and HDL to calculate.

CARDIOVASCULAR

Nordestgaard & Varbo Lancet 2014

Remnant Cholesterol

Residual Lipoprotein CVD Risk

IN PLAIN TERMS This is the cholesterol that standard tests routinely miss. Leftover fat particles (remnants) penetrate artery walls and deposit cholesterol in plaques independently of LDL, making it an important hidden cardiovascular risk factor.

29.8
Scored out of 80

Optimal residual risk

Cholesterol carried in leftover fat particles (IDL, VLDL) after digestion. A proven risk factor for plaque build-up in arteries, independent of your LDL (bad cholesterol). High levels are dangerous even when your standard cholesterol panel looks normal.

Formula: Total Cholesterol − LDL − HDL. Remnant particles directly infiltrate arterial walls to deposit cholesterol in plaques. Requires TC, LDL, and HDL to calculate.

METABOLIC

Matthews et al. Diabetologia 1985

HOMA-IR

Homeostatic Model Assessment of Insulin Resistance

IN PLAIN TERMS This score shows how hard your body works to keep blood sugar under control. When cells stop responding to insulin efficiently, the pancreas overproduces it, quietly driving diabetes risk higher over years before any symptom appears.

3.51
Scored out of 6

Significant insulin resistance

Combines your fasting blood sugar and fasting insulin into one score. High values mean your pancreas is working harder than normal to control blood sugar, which is a sign of insulin resistance.

Formula: (glucose mmol/L × insulin mIU/L) ÷ 22.5. Requires both fasting glucose and fasting insulin. Validated against the euglycemic-hyperinsulinemic clamp, the gold standard for measuring insulin resistance.

METABOLIC

Katz et al. JCEM 2000

QUICKI

Quantitative Insulin Sensitivity Check Index

IN PLAIN TERMS Think of this as the flip side of HOMA-IR. It measures how readily your cells respond to insulin signals. A higher score means better insulin sensitivity, protecting you against diabetes and metabolic disease.

0.32
Scored out of 0.45

Insulin resistance observed

Calculated from your blood sugar and insulin levels. **Higher values are better**, meaning your cells respond well to insulin. Sensitive enough to detect early insulin resistance before HOMA-IR rises.

Formula: 1 ÷ ($\log_{10}$ glucose + $\log_{10}$ insulin). Correlates more linearly with clamp-measured sensitivity than HOMA-IR. Requires fasting insulin for computation.

METABOLIC

Simental-Mendia et al. Eur J Clin Invest 2008

TyG Index

Triglyceride-Glucose Index

IN PLAIN TERMS A clever way to detect insulin resistance without a fasting insulin test. When both your blood sugar and triglycerides are elevated together, it signals a combined metabolic stress that raises your risk of diabetes and heart disease.

4.83
Scored out of 5.2

Higher metabolic risk

An insulin-free way to detect insulin resistance. Combines your fasting triglycerides and blood sugar to measure combined fat-sugar metabolic stress. Higher values are linked to greater risk of diabetes and heart disease, regardless of your BMI.

Formula: ln(TG mg/dL × FPG mg/dL) ÷ 2. Uniquely does not require fasting insulin. Strongly correlates with euglycemic clamp results across diverse populations.

LIVER

Bedogni et al. BMC Gastroenterol 2006

Fatty Liver Index (FLI)

Non-Invasive Hepatic Steatosis Predictor

IN PLAIN TERMS Think of this as a non-invasive liver fat radar. Excess fat in the liver can silently cause inflammation and scarring. This score estimates that risk using your blood results and body measurements, with no scan or biopsy needed.

65.25
Scored out of 100

Elevated — screening index for fatty liver risk (estimated — waist not measured). Not diagnostic.

Uses BMI, waist size, triglycerides, and a liver enzyme (GGT) to screen for fat build-up in the liver without any scan or biopsy. Fatty liver raises the risk of liver scarring, diabetes, and heart disease.

Score ≥60 has ~87% specificity for fatty liver confirmed by ultrasound. Uses a logistic model combining waist circumference, BMI, triglycerides, and GGT, all established metabolic-liver markers.

LIVER

Wai et al. Hepatology 2003

APRI

AST-to-Platelet Ratio Index

IN PLAIN TERMS Your liver releases an enzyme (AST) when its cells are damaged, and platelet counts fall as scarring progresses. This index reads both signals together to detect early liver fibrosis before any symptom or pain appears.

0.45
Scored out of 2

Fibrosis unlikely

Combines a liver enzyme (AST) with your platelet count. Platelet counts fall as liver scarring progresses. Detecting an elevated APRI early allows treatment before irreversible liver damage occurs.

Formula: (AST ÷ 40 U/L) ÷ (platelets × 10^{-9}/L) × 100. Widely used to screen for liver fibrosis without biopsy. Endorsed by WHO for hepatitis C fibrosis staging in low-resource settings.

HOW TO READ THESE INDEX CARDS

Green : Optimal / Low Risk **Orange** : Borderline / Needs Attention **Red** : Elevated Risk / Action Required

Each score is calculated directly from your blood results using validated formulas. A green result means that index is within a healthy range today. **Orange or red scores highlight areas where your body is under measurable stress.** The earlier these are addressed, the better your long-term outcome. Personalised dietary, supplement, and lifestyle recommendations for every flagged index are provided throughout this report. Follow those steps consistently and retest at your next panel to track improvement.

Your Health Summary

Demo User
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What is this page? This is a single, consolidated view of every key finding from your full report: causes, symptoms, and recommended actions, all in one place. It is produced by our **intelligent health system** that compares your actual lab values against healthy reference ranges. Every insight here is directly traceable to your lab results. Nothing is assumed or generalised. Use this page to quickly understand where your health stands today and what steps to take first.

HEALTH ANALYSIS

Consolidated Causes & Symptoms

Your health report reveals several concerning findings that require attention, particularly in your metabolic and inflammatory profiles. Your fasting glucose level is elevated at 105.6 mg/dl, surpassing the normal range of 70.0 - 100.0 mg/dl. Additionally, your HbA1c is at 6.2%, indicating increased risk for diabetes as it exceeds the normal threshold of 4.2 - 5.7%. These results, combined with a HOMA-IR value of 3.51, signify significant insulin resistance, suggesting that your body is struggling to manage blood sugar effectively. Furthermore, your inflammation markers show mild elevation, with an hs-CRP level of 2.2 mg/L and an ESR of 17.0 mm/hr, indicating a potential inflammatory process that could exacerbate other health issues. Your kidney function is generally normal; however, your Urea/Creatinine Ratio is elevated at 34.66, which may suggest variations in hydration or protein intake. In addition, your bone health is mildly affected due to an elevated homocysteine level of 25.0 µmol/L, indicating potential deficiencies in B vitamins. Lastly, while your liver function tests are normal, it is important to maintain a healthy lifestyle to support liver health, as ongoing issues could arise from dietary choices and weight management.

WELLNESS GUIDANCE

Consolidated Holistic Actions

To address your elevated fasting glucose and HbA1c levels, it is crucial to implement immediate dietary changes and lifestyle modifications. Focus on a balanced diet rich in whole foods, including vegetables, whole grains, lean proteins, and healthy fats, while minimizing refined carbohydrates and sugars. Engage in at least 150 minutes of moderate exercise weekly to enhance insulin sensitivity. Regularly monitor your blood glucose levels at home and consult your physician about potential nutritional supplements, such as magnesium and omega-3 fatty acids, to support metabolic health. For your mildly elevated inflammation markers, incorporate anti-inflammatory foods into your diet, such as fatty fish, nuts, seeds, fruits, and vegetables. Maintaining adequate hydration and managing stress through mindfulness practices can further support your overall health. Additionally, ensure a balanced diet rich in B vitamins to address the elevated homocysteine levels, and consider discussing supplementation with your healthcare provider. Schedule follow-up appointments to monitor your metabolic, inflammatory, and kidney health, ensuring that you remain on track with your health goals and adjustments.

THIS MONTH'S FOCUS

Top 3 Priorities

1. Implement dietary changes to lower fasting glucose and HbA1c levels
2. Increase physical activity to enhance insulin sensitivity and manage weight
3. Incorporate anti-inflammatory foods and ensure adequate hydration for overall health

Biological Age

Chronological Age

Your actual age in years since birth, a fixed measure of time that advances identically for everyone regardless of health or lifestyle. It is the **baseline reference** in this report. While it cannot be changed, it sets the benchmark against which your body's true functional condition is compared.

VS

Biological Age

How old your cells and body systems actually function, calculated using the scientifically validated **PhenoAge model** based on nine blood markers. Unlike your calendar age, your biological age can be **improved**: regular exercise, good nutrition, restful sleep, and managing stress can measurably lower it over time.

Chronological Age
Calendar years since birth

DIFFERENCE

-1.9
yrs

Aging Slower

Biological Age
Based on biomarker analysis

Clinical Note · PhenoAge Algorithm (Levine et al., 2018)

Next page discusses each parameter in detail →

Your biological age is calculated using the scientifically validated **PhenoAge** model, which uses nine blood markers: **Albumin, Creatinine, Glucose, C-Reactive Protein (CRP), Lymphocyte %, MCV, RDW, Alkaline Phosphatase (ALP),** and **WBC**. Together, these markers reflect your inflammation levels, organ function, and how fast your cells are ageing, each weighted by how strongly it predicts healthy ageing in large population studies.

What to expect: Biological age close to or below chronological age is healthy. A gap of ≤ 2 yrs older is acceptable; > 5 yrs signals elevated ageing stress.
Can it be improved? Yes. Regular exercise, quality sleep, an anti-inflammatory diet, and managing chronic conditions can measurably lower biological age over time.
*Biological age is a **wellness tracking metric** for monitoring trends over time, not a clinical diagnosis or medical test. Consult your physician before making health decisions.*

Your Metabolic Profile

What are these metrics & how should you use them?

BMI (Body Mass Index) measures your weight relative to your height. Use it as a quick screening guide, not a standalone diagnosis.
Body Fat % measures what proportion of your body is fat. High levels raise the risk of metabolic conditions; very low levels can also affect health.
Ideal Body Weight is a reference target weight for your height and gender to help guide weight management.

Lean Mass / Fat Mass separates your muscle and bone weight from stored fat, helping to tailor your nutrition and exercise plan.
Resting Metabolic Rate (RMR) is the number of calories your body burns at complete rest just to keep you alive. Knowing this helps prevent under-eating on any diet.
Daily Calorie Need (TDEE) is the total number of calories you need each day to maintain your current weight given your activity level. Adjust your intake around this to lose, maintain, or gain weight.

HEIGHT **183.0** cm	WEIGHT **100.0** kg
IDEAL BODY WEIGHT **73.19** kg Over Weight	BODY FAT **31.13**% High
LEAN MASS **68.87** kg Low	FAT MASS **31.13** kg High
RESTING METABOLIC RATE **1820** kcal/day	DAILY CALORIE NEED **2500** kcal/day

BODY MASS INDEX **29.86** Overweight

BIOMARKER ANALYSIS

Factors Affecting Biological Age

Each card shows one of the nine blood markers used in the **PhenoAge model**: **Albumin, Creatinine, Glucose, CRP (inflammation marker), Lymphocytes (immune cells), MCV, RDW, ALP (liver enzyme),** and **WBC (white blood cells)**. The **colour bar** shows where your reading sits within the healthy range; a marker near the centre means you are in the best position. The **chart** maps your result into one of four health zones: **Optimal**, **Mild**, **Moderate**, and **Severe**. If you have had previous tests, earlier readings appear as smaller dots connected by a trend line, showing how each marker has changed over time.

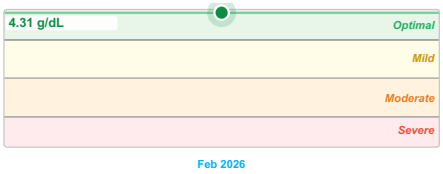
Albumin

4.31 g/dL

Taken: Feb 2026

Within optimal range

3.5 – 5.5



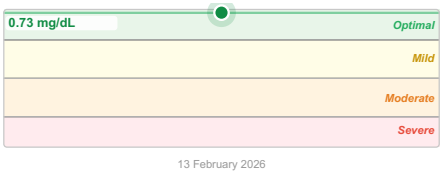
Creatinine

0.73 mg/dL

Taken: 13 February 2026

Within optimal range

0.7 – 1.3



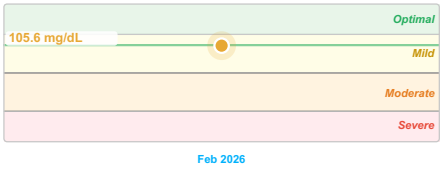
Glucose

105.6 mg/dL

Taken: Feb 2026

Slightly off range

70.0 – 100.0



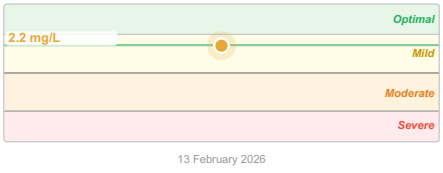
CRP

2.2 mg/L

Taken: 13 February 2026

Moderately elevated

0.0 – 1.0



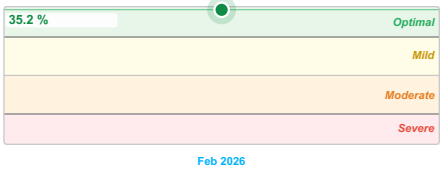
Lymphocytes

35.2 %

Taken: Feb 2026

Within optimal range

20.0 – 40.0



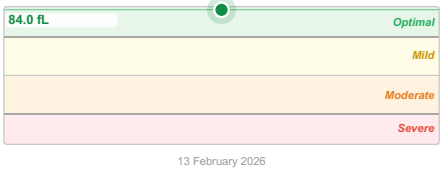
MCV

84.0 fL

Taken: 13 February 2026

Within optimal range

80.0 – 100.0



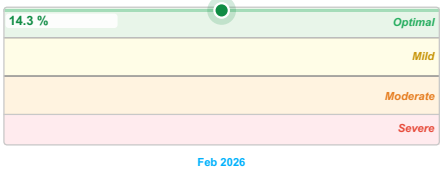
RDW-CV

14.3 %

Taken: Feb 2026

Within optimal range

11.5 – 14.5



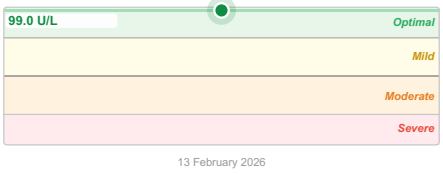
ALP

99.0 U/L

Taken: 13 February 2026

Within optimal range

40.0 – 120.0



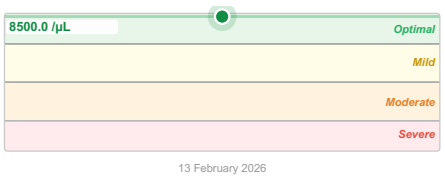
WBC

8500.0 /µL

Taken: 13 February 2026

Within optimal range

4000.0 – 10000.0



ORGAN HEALTH ASSESSMENT

Organ Health Scorecard

Each score **(0–100)** represents the functional health of an organ system derived from your latest blood test results. Higher scores indicate better function. Colour bands reflect risk level: **green** = optimal, **amber** = mild-to-moderate risk, **red** = high risk. Use this page as an at-a-glance health snapshot and discuss any flagged systems with your doctor or health advisor.

Diabetic Profile

Your metabolic health indicates how well your body processes nutrients and regulates blood sugar. It's a key indicator of

MODERATE RISK

50 /100

Moderate

KEY FACTORS

- HbA1c — Deranged
- Fasting Glucose — Deranged
- HOMA-IR — Deranged

Liver Function

The liver is your body's detoxification powerhouse, processing nutrients and filtering toxins. Liver health is essential for overall metabolic

OPTIMAL

100 /100

Normal

KEY FACTORS

✓ All parameters within normal range

Lipid Profile

Heart health reflects your cardiovascular system's efficiency, including cholesterol levels and lipid balance. Good heart health reduces risk of

MILD RISK

75 /100

Mild

KEY FACTORS

- hs-CRP — Deranged

Kidney Function

Kidneys filter waste from your blood and regulate fluid balance. Kidney function is critical for blood pressure control and waste elimination.

OPTIMAL

99 /100

Normal

KEY FACTORS

- Urea / Creatinine Ratio — Deranged

Thyroid Profile

The thyroid gland regulates metabolism, energy, and body temperature through hormone production. Thyroid balance is essential for vitality.

OPTIMAL

100 /100

Normal

KEY FACTORS

✓ All parameters within normal range

Inflammation

Chronic inflammation is linked to many diseases. Lower inflammation levels indicate better immune function and reduced disease risk.

MILD RISK

75 /100

Mild

KEY FACTORS

- hs-CRP — Deranged
- ESR — Deranged

Vitamin Profile

Bone health depends on calcium, vitamin D, and other nutrients. Strong bones prevent fractures and support overall structural health.

MILD RISK

75 /100

Mild

KEY FACTORS

- Homocysteine — Deranged
- Vitamin D — Deranged

Hemogram

Blood health encompasses red cells, white cells, and platelets. Optimal blood parameters ensure oxygen delivery and immune function.

OPTIMAL

99 /100

Normal

KEY FACTORS

- MPV — Deranged

SCORE LEGEND — Optimal · Moderate Risk · High Risk · No Data

RECOMMENDED NUTRITION PATTERN

Nutrition & Macro Profile

Your macro-distribution (the balance of carbohydrates, protein, and fat in your diet) has been calculated from your Basal Metabolic Rate (how many calories you burn at rest), your activity level, and your blood test results. Use these percentages as a **daily calorie guide**: prioritise hitting your protein target first, then distribute carbohydrates and healthy fats accordingly. Sticking to this consistently for 4–6 weeks produces measurable improvements in energy levels, body composition, and blood markers.

A personalised 14-Day Meal Plan built around these exact macro targets, including daily recipes, is included at the last pages of this report.

BASAL METABOLIC RATE

1820

KCAL / DAY

Calories your body needs at complete rest to sustain essential functions.

CARBOHYDRATES

30%

OF DAILY CALORIES

Primary energy source fuelling brain function and physical activity.

PROTEINS

35%

OF DAILY CALORIES

Supports tissue repair, muscle maintenance, and enzyme production.

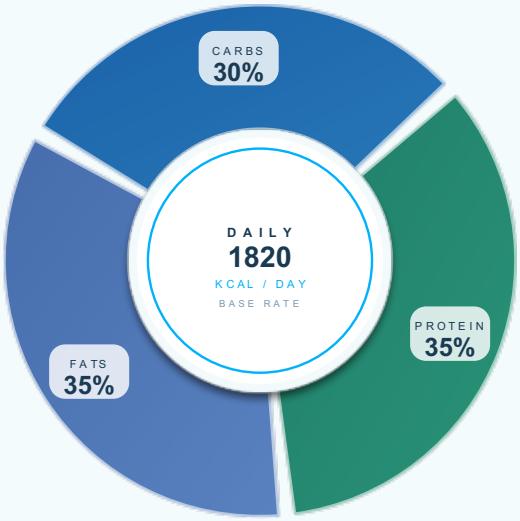
HEALTHY FATS

35%

OF DAILY CALORIES

Promotes hormone balance, brain health, and fat-soluble vitamin absorption.

Your diet preference is **Non-Vegetarian*** **based on your given input*



Carbs 30% | Protein 35% | Healthy Fats 35%

YOUR RECOMMENDED NUTRITION PATTERN

Low Glycaemic Diet Pattern

Designed to keep your blood sugar steady by choosing slow-digesting carbohydrates, enough fibre, and balanced protein. Supports healthy insulin response, stable energy, and weight management.

Consistency is the cornerstone of results: aim to follow your pattern for at least 4–6 weeks to see meaningful changes in energy, weight, and blood markers.

If you lapse, don't reset the clock. Simply **resume from your next meal.**One off-day does not undo progress; sustained consistency over time is what drives lasting improvement.

RECOMMENDED NUTRITION PATTERN

Your Superfoods & Lifestyle Biohacks

Your **Superfoods & Lifestyle Biohacks** are chosen based on your blood test results and organ health scores. Each recommendation targets your specific nutritional needs and lifestyle risk factors. Incorporate these consistently alongside your meal plan for best results. The **number on each item** shows how many of your conditions it addresses, with a higher number meaning greater relevance to your health profile.

★ Your Superfoods

Beans / Lentils 9

+ Low-glycaemic plant protein with soluble fibre that stabilises blood sugar
+ Rich in folate, iron and zinc supporting red blood cell production
+ Resistant starch feeds gut microbiome and improves insulin sensitivity

Lowers fasting glucose · Improves HbA1c control

Oats 9

+ Beta-glucan soluble fibre lowers LDL cholesterol and slows glucose absorption
+ Prebiotic fibre feeds beneficial gut bacteria and improves microbiome diversity
+ Rich in manganese and B vitamins that support energy metabolism

Lowers fasting glucose · Improves HbA1c control

Chia seeds 8

+ Rich in omega-3 ALA and soluble fibre to reduce triglycerides and stabilise blood glucose
+ High calcium and phosphorus content supports bone density and skeletal health
+ Complete amino acid profile provides all essential proteins for tissue repair

Lowers hs-CRP inflammation · Lowers fasting glucose

Cinnamon 8

+ Moderates post-meal blood glucose by slowing carbohydrate digestion
+ May improve insulin sensitivity by enhancing glucose transporter activity
+ Cinnamaldehyde provides anti-inflammatory and antioxidant cellular protection

Lowers fasting glucose · Improves HbA1c control

Berries 9

+ Anthocyanins and polyphenols reduce oxidative stress and lower CRP
+ Low-glycaemic fruit with high vitamin C supporting immune defence
+ Pterostilbene and resveratrol support cardiovascular and brain health

Lowers hs-CRP inflammation · Improves HbA1c control

Foxtail millet 6

+ Low-glycaemic grain with fibre and magnesium stabilising blood sugar levels
+ Rich in iron and B vitamins supporting energy metabolism and haemoglobin
+ Silica content supports bone density and connective tissue strength

Lowers fasting glucose · Improves HbA1c control

Kodo millet 7

+ Low-GI grain with polyphenol antioxidants showing anti-inflammatory properties
+ Rich in flavonoids and tannins that inhibit alpha-amylase enzyme activity
+ High fibre content slows gastric emptying and reduces post-meal glucose spikes

Lowers hs-CRP inflammation · Lowers fasting glucose

Moringa 9

+ Rich in non-heme iron and anti-inflammatory antioxidants for blood health
+ Isothiocyanates activate detoxification enzymes and reduce oxidative stress
+ Chlorogenic acid slows glucose absorption and supports glycaemic control

Boosts iron absorption · Lowers hs-CRP inflammation

Leafy greens 9

+ Folate, vitamin K and chlorophyll antioxidants support liver health
+ Vitamin K2 supports calcium regulation and prevents arterial calcification
+ Nitrate content converts to nitric oxide improving blood flow and pressure

Lowers hs-CRP inflammation · Boosts iron absorption

Spinach 7

+ Excellent source of non-heme iron and folate for healthy red blood cell formation
+ High vitamin K1 supports blood clotting and bone mineralisation
+ Lutein and zeaxanthin carotenoids protect against macular degeneration

Boosts iron absorption · Lowers hs-CRP inflammation

▼ Your Lifestyle Biohacks

Resistance training (2-3 sessions/week, compound exercises)

Building lean muscle increases baseline glucose disposal — each extra kg of muscle burns ~…

2-3 sets × 8-12 reps, 6-8 compound exercises · Measurable HbA1c and HDL improvement in 8-12 weeks

Time-restricted eating (14:10 or 16:8 eating window)

Compressing eating to a 8-10 hour window lowers fasting insulin, reduces liver fat accumul…

14-16 hour overnight fast · Visible improvements in 4-6 weeks

Consistent 7-9h sleep (fixed bedtime and wake time every day)

Chronic sleep deprivation raises cortisol by 37%, increases insulin resistance and elevate…

7-9 hours per night · Metabolic improvements in 2-4 weeks

HIIT — High Intensity Interval Training (2×/week)

HIIT generates the largest improvements in insulin sensitivity per minute of exercise, red…

20-30 min: 10× 1-min intervals at 85-95% max HR with 1-min active recovery · HbA1c and HDL improvements in 8-12 weeks

Morning sunlight exposure (10-30 min within 1 hour of waking)

Morning bright light synchronises the circadian clock, improves sleep quality, regulates c…

10-30 min of natural outdoor light (face and arms exposed) · Circadian benefits in 1-2 weeks; Vitamin D in 4-8 weeks

Zone 2 aerobic cardio (brisk walk, jog, or cycle at 60-70% max HR)

Zone 2 intensity primarily burns fat for fuel, directly lowering triglycerides and LDL whi…

30-45 min per session · Lipid improvements in 4-8 weeks

Mindfulness or MBSR meditation (15-20 min/day)

Mindfulness-based stress reduction lowers cortisol by 20-30%, reducing cortisol-driven ins…

15-20 min session · CRP reductions measurable in 8 weeks

Sedentary break protocol (2-min walk every 30 min of sitting)

Breaking prolonged sitting with 2-minute walks every 30 minutes reduces postprandial gluco…

2 min walk or light movement · Immediate postprandial glucose and triglyceride benefits

Daily fermented foods or probiotic supplementation

Regular consumption of fermented foods (curd, kefir, kimchi) or probiotics restores gut mi…

150-200g fermented food/day OR 10-50 billion CFU probiotic supplement · Microbiome diversity shifts in 3-4 weeks; inflammatory markers in 6-8 weeks

Cold shower or cold water immersion (2-5 min at 14-18°C)

Cold exposure activates brown adipose tissue (BAT), increases norepinephrine by 300%, supp…

2-5 min cold exposure (start with 30s, gradually increase) · 2-4 weeks for physiological adaptation

RECOMMENDED NUTRITION PATTERN

Your Essential Supplements

Your **Essential Supplements** have been selected based on your individual blood test results and organ health scores. Each supplement directly addresses the specific deficiencies or imbalances identified in your lab data. Start with dietary changes first and introduce supplements one at a time.

⚠ **Important:** These are general wellness suggestions based on your lab results. Consult your physician or pharmacist before starting any supplement, especially if you are pregnant, nursing, have a chronic condition, or are currently on any medication. This report is not a prescription or substitute for medical advice. You do not need to start all supplements together. Prioritise dietary changes first and introduce supplements one at a time, under professional guidance.

Magnesium

Cofactor for 300+ enzymatic reactions; improves insulin sensitivity, lowers fasting glucose, HbA1c and triglycerides, reduces blood pressure, relieves muscle cramps, supports bone density and lowers systemic inflammation

Dose: 200–400 mg/day (glycinate or citrate)
Duration: Long-term
⚠ Reduce dose if loose stools occur; caution in severe kidney disease

Lowers fasting glucose | Improves HbA1c control

Berberine

Activates AMPK (the body's metabolic switch) to lower fasting glucose, HbA1c, LDL cholesterol and triglycerides; reduces elevated liver enzymes; clinical efficacy comparable to metformin for glucose control

Dose: 500 mg 2–3×/day with meals
Duration: 12 weeks, then reassess
⚠ Interacts with metformin and blood pressure medications; not for pregnant or nursing women

Lowers fasting glucose | Improves HbA1c control

Alpha Lipoic Acid

Universal antioxidant (water and fat soluble) that improves insulin sensitivity, lowers fasting glucose and HbA1c, reduces oxidative stress, lowers elevated ALT and supports nerve health in diabetic neuropathy

Dose: 300–600 mg/day
Duration: 8–12 weeks
⚠ May lower blood sugar; caution if on hypoglycaemic medication

Lowers fasting glucose | Improves HbA1c control

Ashwagandha

Adaptogenic herb that measurably reduces elevated cortisol (KSM-66 trials: −28%), improves thyroid T3/T4, lowers cortisol-driven blood glucose, reduces hs-CRP and supports testosterone in males

Dose: 300–600 mg/day (standardised extract)
Duration: 8–12 weeks
⚠ Caution in thyroid disorders and autoimmune disease; not for pregnant women

Lowers fasting glucose | Lowers hs-CRP inflammation

Vitamin D3

Corrects Vitamin D deficiency to restore bone mineralisation, calcium absorption, immune function and muscle strength; reduces inflammatory CRP and lowers fracture risk

Dose: 1000–4000 IU/day
Duration: 12 weeks, recheck levels
⚠ Toxicity risk above 10,000 IU/day; check Vitamin D level before increasing dose

Corrects Vitamin D deficiency | Lowers hs-CRP inflammation

TMG (Betaine)

Donates methyl groups to convert homocysteine to methionine (betaine-homocysteine methyltransferase pathway), significantly reducing elevated homocysteine, supporting glucose metabolism and reducing inflammation

Dose: 500–2000 mg/day
Duration: 3–6 months
⚠ May cause GI discomfort; high doses may raise LDL in some individuals

Reduces elevated homocysteine | Lowers fasting glucose

Zinc

Supports immune defence, raises HDL cholesterol, improves insulin signalling, aids haemoglobin production, activates antioxidant enzymes (SOD) and is essential for testosterone synthesis via 5-alpha reductase activity

Dose: 15–30 mg/day
Duration: 3–6 months, then reassess
⚠ Long-term use above 40 mg/day depletes copper; take with food to reduce nausea

Lowers hs-CRP inflammation | Supports testosterone production

Probiotics

Restores gut microbiome diversity and tight junction integrity, reduces intestinal permeability and systemic inflammation (hs-CRP), improves insulin sensitivity and lowers fasting glucose via gut-metabolic axis

Dose: 10–50 billion CFU/day (multi-strain)
Duration: 4–12 weeks, then reassess
⚠ Caution in immunocompromised individuals

Lowers hs-CRP inflammation | Lowers fasting glucose

★ Recommended for Overall Wellness

Widely endorsed by WHO, NIH & major health authorities for general wellness. Not included in your personalised list above. These may be considered if advised by a clinician.

Omega-3 (EPA+DHA)

EPA and DHA significantly lower triglycerides (by 20–50%), raise HDL and ApoA1, reduce hs-CRP and ApoB, suppress NF-κB inflammatory pathways and reduce cardiovascular event risk

Dose: 2–4 g EPA+DHA/day
Duration: Long-term

Vitamin C

Significantly enhances non-heme iron absorption, prevents LDL oxidation, reduces inflammatory hs-CRP, lowers oxidative stress and supports immune defence as a potent antioxidant

Dose: 500–1000 mg/day
Duration: Long-term

Vitamin B12

Corrects B12 deficiency to restore nerve conduction, red blood cell production, energy metabolism and reduce elevated homocysteine cardiovascular risk

Dose: 500–1000 mcg/day (methylcobalamin preferred)
Duration: Ongoing if deficient

Ginkgo Biloba

Improves cerebral and peripheral blood flow through PAF antagonism, acts as antioxidant to protect neurons and supports memory, concentration and cognitive performance

Dose: 120–240 mg/day (standardised extract)
Duration: 12 weeks

RECOMMENDED NUTRITION PATTERN

Grains & Oils Guide

Your **Grains & Oils Guide** is personalised based on your blood glucose, lipid panel, and inflammatory markers. Selecting the right grains supports stable blood sugar and gut health, while optimal oils promote cardiovascular and anti-inflammatory benefits. **Green** items are recommended for your profile; **red** items should be avoided to align with your health goals.

RECOMMENDED

Grains You Should Consume

EAT

Foxtail millet
Lowers fasting glucose, Improves HbA1c control, Boosts iron absorption, Lowers hs-CRP inflammation

Pearl millet (Bajra)
Boosts iron absorption, Lowers fasting glucose, Lowers hs-CRP inflammation, Improves HbA1c control

Teff
Boosts iron absorption, Lowers fasting glucose, Lowers hs-CRP inflammation, Improves HbA1c control

Oats
Lowers fasting glucose, Improves HbA1c control, Lowers hs-CRP inflammation

Barley
Lowers fasting glucose, Improves HbA1c control, Lowers hs-CRP inflammation

Quinoa
Lowers fasting glucose, Lowers hs-CRP inflammation, Improves HbA1c control

Buckwheat
Lowers fasting glucose, Lowers hs-CRP inflammation, Improves HbA1c control

Whole wheat
Lowers fasting glucose, Lowers hs-CRP inflammation, Improves HbA1c control

Brown rice
Lowers fasting glucose, Improves HbA1c control, Lowers hs-CRP inflammation

AVOID

Grains You Should Avoid

SKIP

Maida
Spikes blood glucose, Worsens long-term glucose control, Promotes systemic inflammation

White rice
Spikes blood glucose, Worsens long-term glucose control, Promotes systemic inflammation

Sugary cereals
Spikes blood glucose, Worsens long-term glucose control, Promotes systemic inflammation

White bread
Spikes blood glucose, Worsens long-term glucose control, Promotes systemic inflammation

Rava
Spikes blood glucose, Worsens long-term glucose control, Promotes systemic inflammation

Rice flour
Spikes blood glucose, Worsens long-term glucose control, Promotes systemic inflammation

SMART OIL GUIDE

- **Choose cold-pressed oils:** cold pressing preserves heat-sensitive omega-3s, polyphenols, and vitamin E lost during refining.
- **Match oil to heat:** use mustard, rice bran, or groundnut oil for high-heat cooking; reserve EVOO, flaxseed, and walnut oil for dressings and drizzles.
- **Avoid reusing or overheating oils:** reheated oils form aldehydes and trans fats that worsen inflammation and cardiovascular risk.

RECOMMENDED

Oils You May Consume

USE

EVOO
Lowers hs-CRP inflammation, Lowers fasting glucose

Rice bran oil
Lowers fasting glucose, Lowers hs-CRP inflammation

Sesame oil
Lowers hs-CRP inflammation, Lowers fasting glucose

Walnut oil
Lowers hs-CRP inflammation, Lowers fasting glucose

Almond oil
Lowers hs-CRP inflammation, Lowers fasting glucose

Flaxseed oil
Lowers hs-CRP inflammation

AVOID

Oils You Should Avoid

AVOID

Palm oil
Promotes systemic inflammation

Vanaspati
Promotes systemic inflammation

Margarine (stick)
Promotes systemic inflammation

Soybean oil
Promotes systemic inflammation

Corn oil
Promotes systemic inflammation

Coconut oil
Promotes systemic inflammation

Your Exercise Prescription

Personalized Movement & Exercise Protocol

Demo's metabolic (blood glucose/insulin) markers indicate moderate risk. With a BMI of 29.9 and a lightly active activity level, these targeted exercise prescriptions are selected to address the underlying physiological pathways, with safety precautions noted where applicable. Protocols are ranked by clinical relevance to your biomarker profile, drawn from peer-reviewed exercise medicine literature.

Exercise Protocol

CARDIO

Zone 2 Aerobic Walking

45–60 min/session | ▶ 5 days/week | ★ Low–Moderate (Zone 2, 60–70% max HR)

ADDRESSES **Metabolic** · **Cardiovascular** · **Inflammation** · **Weight** ·

WHY IT HELPS?

Zone 2 training maximises mitochondrial biogenesis and fat oxidation by keeping the heart rate in a range where fat is the primary fuel source. Sustained low-intensity aerobic work lowers fasting insulin, reduces HOMA-IR, and promotes systemic anti-inflammatory adaptations via reduced visceral adiposity.

HOW IT BENEFITS YOU?

- ✓ Reduces fasting insulin and HOMA-IR by up to 25%
- ✓ Lowers LDL and triglycerides via improved lipid oxidation
- ✓ Reduces systemic CRP and inflammatory cytokines

Ref *Holloszy & Coyle, J Appl Physiol 1984; Hawley & Lessard, J Physiol 2008*

ⓘ **For You** At age 50 your Zone 2 target HR is 102–119 bpm (est. max HR 170 bpm).

HIIT

HIIT – High-Intensity Interval Training

20–30 min/session | ▶ 3 days/week (non-consecutive) | ★ High (85–95% max HR during work intervals)

ADDRESSES **Metabolic** · **Cardiovascular** · **Weight** ·

WHY IT HELPS?

HIIT produces rapid GLUT4 upregulation and catecholamine-driven lipolysis in a fraction of the time required by steady-state cardio. Short bursts of high effort trigger excess post-exercise oxygen consumption (EPOC), elevating caloric burn for 12–24 hours after the session.

HOW IT BENEFITS YOU?

- ✓ Superior HOMA-IR reduction vs steady-state cardio in same time
- ✓ Significant triglyceride reduction via catecholamine-driven lipolysis
- ✓ Improves VO2 max by 15–20% within 6 weeks

Ref *Gibala et al., J Physiol 2012; Little et al., J Appl Physiol 2011*

ⓘ **For You** At age 50 your HIIT work-interval HR target is 144–162 bpm. At age 50, warm up for a full 10 minutes before high-intensity work and allow 48 hours recovery between sessions.

⚠ Avoid if severe cardiovascular or kidney dysfunction is present. Consult physician before starting.

STRENGTH

Resistance / Strength Training

45 min/session | ▶ 3 days/week | ★ Moderate (60–75% 1RM, compound lifts)

ADDRESSES **Metabolic** · **Weight** · **Bone Health** · **Thyroid** ·

WHY IT HELPS?

Resistance training upregulates GLUT4 transporter expression in skeletal muscle, improving insulin-mediated glucose uptake independent of aerobic exercise. It preserves and builds lean mass - raising resting metabolic rate - and applies mechanical load to bone to directly stimulate osteogenesis.

HOW IT BENEFITS YOU?

- ✓ Increases insulin sensitivity via GLUT4 upregulation in muscle
- ✓ Preserves and builds lean body mass, raising BMR
- ✓ Stimulates bone mineral density via mechanical loading

Ref *Westcott WL, Curr Sports Med Rep 2012; Strasser & Schobersberger, J Obes 2011*

CARDIO

Cycling (Stationary or Outdoor)

40–60 min/session | ▶ 4 days/week | ★ Moderate (60–75% max HR, 70–90 RPM)

ADDRESSES **Cardiovascular** · **Weight** · **Metabolic** ·

WHY IT HELPS?

Cycling provides sustained moderate-intensity aerobic stimulus that lowers LDL, raises HDL, and reduces triglycerides through improved lipoprotein lipase activity. Non-weight-bearing nature makes it ideal for obese or arthritic patients.

HOW IT BENEFITS YOU?

- ✓ Reduces LDL cholesterol and raises HDL via lipoprotein lipase
- ✓ Lowers resting heart rate and blood pressure sustainably
- ✓ Burns 400–600 kcal/hour with low joint loading

Ref *Myers et al., NEJM 2002; Poirier et al., Circulation 2006*

ⓘ **For You** At age 50 your target HR for this exercise is 102–128 bpm.

CARDIO

Endurance Running / Jogging

30–45 min/session | ▶ 3–4 days/week | ★ Moderate–High (70–80% max HR)

ADDRESSES **Cardiovascular** · **Metabolic** · **Weight** ·

WHY IT HELPS?

Sustained jogging at moderate-high intensity is one of the most effective interventions for reducing cardiovascular risk, lowering triglycerides, and increasing HDL. Running distance is inversely correlated with all-cause mortality and cardiovascular events in large epidemiological studies.

HOW IT BENEFITS YOU?

- ✓ Reduces cardiovascular mortality risk by 30–45% vs sedentary controls
- ✓ Raises HDL and lowers triglycerides significantly
- ✓ Burns 600–800 kcal/hour for weight management

Ref *Williams PT & Thompson PD, PLOS ONE 2013; Lee et al., J Am Coll Cardiol 2014*

ⓘ **For You** At age 50 your jogging target HR is 119–136 bpm. At age 50, warm up for a full 10 minutes before high-intensity work and allow 48 hours recovery between sessions.

⚠ Not recommended if severe cardiovascular disease is present. Build up gradually.

STRENGTH

Functional Strength Circuit

30–40 min/session | ▶ 3 days/week | ★ Moderate (bodyweight or light-moderate loads)

ADDRESSES **Metabolic** · **Weight** · **Bone Health** ·

WHY IT HELPS?

Circuit-style functional training combines cardiovascular stimulus with multi-joint strength movements, producing superior glucose disposal and EPOC compared to isolated machine exercises. Weight-bearing movements directly stimulate osteoblast activity for bone formation.

HOW IT BENEFITS YOU?

- ✓ Simultaneous cardio and strength benefit in 30–40 min sessions
- ✓ Improves multi-joint functional movement and metabolic flexibility
- ✓ Stimulates bone mineral density through axial and ground-reaction loading

Ref *Garber et al., Med Sci Sports Exerc 2011; Westcott et al., J Strength Cond Res 2012*

WEEKLY EXERCISE TARGET

Weekly targets: 150+ min moderate aerobic activity, plus 2–3 strength sessions.

11h 9m+	6–7	Daily	9,661
~96 min/day	Sessions	Practice	Steps
Aerobic / Week	Strength / Week	Mind-Body	Est. / Day

WHO Physical Activity Guidelines: Adults should accumulate **150–300 minutes** of moderate-intensity aerobic activity per week (or 75–150 minutes vigorous-intensity), plus muscle-strengthening activities on **2 or more days** per week. Reducing sedentary time and including balance & flexibility training further lowers the risk of chronic disease.

Diabetic Profile

This section shows your **Metabolic Profile** based on your blood test results. Your blood sugar control, how your body responds to insulin, and related markers are compared against healthy reference ranges for your age and sex. Values shown in **red** fall outside the healthy range and may be an early sign of metabolic issues; review the implications and recommended actions below to take steps towards better metabolic health.

Clinical Implications

Your metabolic profile indicates moderate severity, with several abnormal markers. Your fasting glucose level is 105.6 mg/dl, which is high compared to the reference range of 70.0 - 100.0 mg/dl. Your HbA1c is 6.2%, also high, indicating increased diabetes risk, as it exceeds the normal range of 4.2 - 5.7%. Additionally, your HOMA-IR value of 3.51 signifies moderate to significant insulin resistance, with a normal range of 0 - 1.9. These findings collectively suggest that your body is struggling to regulate blood sugar effectively, which could lead to further metabolic complications if not addressed.

Recommended Action

To address your metabolic profile, prioritize dietary changes and lifestyle modifications. Focus on a balanced diet rich in whole foods, including vegetables, whole grains, lean proteins, and healthy fats. Limit refined carbohydrates and sugars, which can exacerbate insulin resistance. Regular physical activity, such as at least 150 minutes of moderate exercise weekly, is crucial for improving insulin sensitivity. Consider monitoring your blood glucose levels regularly, ideally at home, to track your progress. Additionally, discuss with your physician about possible nutritional supplements, such as magnesium and omega-3 fatty acids, which may support metabolic health. Regular follow-ups are recommended to reassess your HbA1c and fasting glucose levels.

HOMA-IR: 3.51 - Significant insulin resistance

DEPARTMENT OF BIOCHEMISTRY			
Test Name	**Value**	**Unit**	**Bio. Ref Interval**
INSULIN (FASTING)	**13.48**	mU/L	3 - 25
Glucose, Fasting	**105.6**	mg/dl	70 - 100
Average Estimated Glucose - plasma	**131.24**		70 - 126
Hba1c (Glycosylated Hemoglobin)	**6.2**	%	4.2 - 5.7

Lipid Profile

This section presents your **Lipid & Cardiac Profile** based on your personal blood lipid panel. Your cholesterol fractions, atherogenic ratios, and cardiovascular risk markers are evaluated against evidence-based reference thresholds tailored to your profile. Values shown in **red** exceed optimal limits and may contribute to increased cardiovascular risk; follow the recommended actions below to guide your dietary choices and lifestyle modifications for a healthier heart.

Clinical Implications

Your lipid profile indicates a mildly affected cardiovascular status. The high sensitivity C-reactive protein (HsCRP) level is 2.2 mg/L, which is above the normal reference range of 0.0 - 1.0 mg/L, indicating mildly to moderately elevated inflammation and representing an additional cardiovascular risk factor. Your serum HDL cholesterol is 45.1 mg/dL, which is within the optimal range of 40.0 - 60.0 mg/dL. Additionally, your serum triglycerides are at 149.0 mg/dL, which is optimal as well, falling within the reference range of 0.0 - 150.0 mg/dL. The elevated HsCRP warrants attention as it suggests underlying inflammation that could contribute to cardiovascular issues.

Recommended Action

To address your mildly elevated HsCRP, consider incorporating anti-inflammatory foods into your diet, such as fatty fish (like salmon), nuts, seeds, fruits, and vegetables. Regular physical activity, aiming for at least 150 minutes of moderate exercise weekly, can also help reduce inflammation. Monitoring your HsCRP levels every 3 to 6 months is advisable to assess progress. Additionally, ensure adequate hydration and consider discussing omega-3 fatty acid supplementation with your physician. Maintaining a heart-healthy diet, rich in whole grains and low in saturated fats, can further support your cardiovascular health. Regular follow-ups with your healthcare provider are essential to evaluate your lipid profile and inflammation markers.

DEPARTMENT OF BIOCHEMISTRY

Test Name	Value	Unit	Bio. Ref Interval
Serum HDL Cholesterol	**45.1**	mg/dl	40 - 60
Serum Triglycerides	**149**	mg/dl	0 - 150
Total Cholesterol	**177**	mg/dl	0 - 200
Non-HDL Cholesterol	**131.9**	mg/dl	0 - 160.0
Total CHOL / HDL Cholesterol Ratio	**3.92**	Ratio	3.30 - 4.40
HS-CRP (HIGH SENSITIVITY C-REACTIVE PROTEIN)	**2.2**	mg/L	0 - 1.0

Liver Function

This section presents your **Liver Function Profile**, evaluating key hepatic enzymes and proteins that reflect the health and metabolic processing capacity of your liver. ALT, AST, GGT, and ALP assess enzyme activity and can detect hepatocellular damage, biliary obstruction, or alcohol-related liver stress. Bilirubin and albumin levels indicate the liver's excretory function and synthetic reserve. Values shown in **red** fall outside optimal limits and may warrant clinical evaluation and targeted dietary intervention.

Clinical Implications

Your liver function tests are within normal ranges, confirming healthy hepatic enzyme activity and synthetic capacity. ALT, AST, GGT, and ALP levels show no signs of hepatocellular damage or biliary obstruction. Bilirubin and albumin within reference ranges reflect adequate excretory function and protein synthesis.

Recommended Action

Limit alcohol intake to protect hepatic health, as regular consumption elevates liver enzymes over time. Eat a diet rich in antioxidants, including leafy greens, cruciferous vegetables, and berries, to support liver detoxification pathways. Maintain a healthy body weight, as excess visceral fat is the leading driver of non-alcoholic fatty liver disease. Recheck liver enzymes annually.

FLI: 65.25 - Elevated — screening index for fatty liver risk (estimated — waist not measured). Not diagnostic.

DEPARTMENT OF BIOCHEMISTRY			
Test Name	**Value**	**Unit**	**Bio. Ref Interval**
Serum Bilirubin, (Indirect)	**0.44**	mg/dl	0.0 - 0.8
Alkaline Phosphatase (ALP)	**99**	U/L	43 - 115
Serum Total Protein	**7.32**	g/dl	6.6 - 8.3
Alanine Aminotransferase (ALT/SGPT)	**40**	U/L	3 - 50
Serum Globulin	**3.01**	gm/dl	3.0 - 4.2
Gamma Glutamyl Transferase (GGT)	**36**	U/L	5 - 55
Serum Bilirubin, (Direct)	**0.09**	mg/dl	0 - 0.2
Aspartate Aminotransferase (AST/SGOT)	**28**	U/L	3 - 50
Serum Bilirubin, (Total)	**0.53**	mg/dl	0.3 - 1.2
SGOT/SGPT Ratio	**0.7**	Ratio	0.7 - 1.4
Albumin/Globulin Ratio	**1.43**	Ratio	1.2 - 2.5
Serum Albumin	**4.31**	g/dl	3.5 - 5.2

Kidney Function

This section presents your **Kidney Function Profile**, assessing the efficiency of your renal filtration system and its capacity to clear metabolic waste products. Serum creatinine, blood urea, uric acid, and eGFR reflect glomerular filtration capacity and nitrogen metabolism. Elevated values may indicate reduced kidney clearance, dehydration, or early renal stress. Values shown in **red** fall outside optimal limits and require physician review alongside dietary and lifestyle modifications.

Clinical Implications

Your kidney function test results indicate that the overall severity of your kidney health is normal. However, your Urea/Creatinine Ratio is elevated at 34.66, with a reference range of 10.0 - 20.0. This is worth monitoring at your next check-up, as it may suggest variations in hydration or protein intake. On a positive note, your Serum Creatinine level is 0.73 mg/dl, which is within the optimal range of 0.7 - 1.3 mg/dl. Additionally, your Blood Urea Nitrogen (BUN) at 11.8 mg/dl and your estimated GFR of 110.6 mL/min/1.73m² are both optimal, indicating excellent kidney function. Overall, these results reflect a reassuring status for your kidney health.

Recommended Action

To maintain your kidney health, it is important to ensure adequate hydration. Aim to drink enough fluids throughout the day unless otherwise advised by your physician. You may also consider a balanced diet rich in fruits, vegetables, whole grains, and lean proteins to support overall kidney function. Regular physical activity is beneficial, so try to engage in moderate exercise several times a week. It's a good idea to schedule your next kidney function check-up to monitor the Urea/Creatinine Ratio and discuss any dietary adjustments with your healthcare provider. Remember, maintaining a healthy lifestyle is key to preserving your kidney health.

DEPARTMENT OF BIOCHEMISTRY

Test Name	Value	Unit	Bio. Ref Interval
Serum Calcium	9.5	mg/dl	8.8 - 10.6
HOMOCYSTEINE	25	µmol/L	3.7 - 13.9
Serum Uric Acid	6.1	mg/dl	3.5 - 7.2
Urea/Creatinine Ratio	34.66	Ratio	10 - 20
Serum Chloride	103	mmol/L	101 - 109
Blood Urea	25.3	mg/dl	17 - 43
Serum Creatinine	0.73	mg/dl	0.7 - 1.3
Blood Urea Nitrogen (BUN)	11.8	mg/dl	8 - 20
Serum Sodium	142	mmol/L	136 - 146
Bun/Creatinine Ratio	16.19	Ratio	12.1 - 20.1
GFR, ESTIMATED	110.6	mL/min/1.73m2	

i Clinical Note on Creatinine: Low creatinine levels typically indicate low muscle mass, vegetarian diet, or high kidney clearance rate. Unlike elevated creatinine, low creatinine is **not a marker of kidney disease** or impaired kidney function. Only high creatinine levels signal reduced kidney filtration capacity.

Hemogram

This section presents your **Complete Blood Count (Hemogram)**, a comprehensive panel evaluating the cellular components of your blood. Red cell indices reflect oxygen-carrying capacity and iron status; white cell counts assess immune surveillance and inflammatory response; platelet parameters indicate clotting potential. Values shown in **red** signal deviations from normal hematological ranges that may indicate anemia, infection, inflammation, or other conditions requiring clinical follow-up.

Clinical Implications

Your blood parameters indicate a generally healthy status, with one notable observation. The Mean Platelet Volume (MPV) is recorded at 11.2 fL, which is above the reference range of 7.0 - 9.0 fL. While this elevation is worth monitoring at your next check-up, it is important to note that other parameters, including your Absolute Neutrophil Count (ANC) at 2.93 x 10^3/uL, Hematocrit at 41.0%, and Hemoglobin at 13.7 g/dL, are all within optimal ranges. However, your Serum Iron level is at 65.0 ug/dl, slightly below the normal range of 70.0 - 180.0 ug/dl, indicating a low level that may warrant attention in future evaluations.

Recommended Action

To support your overall blood health, consider incorporating iron-rich foods into your diet, such as lean meats, beans, lentils, and leafy green vegetables. Additionally, vitamin C-rich foods like citrus fruits can enhance iron absorption. It is also advisable to maintain a balanced diet and stay hydrated. Regular physical activity can contribute positively to your blood health; aim for at least 150 minutes of moderate exercise weekly. Schedule a follow-up appointment to monitor your MPV and Serum Iron levels, ensuring they remain within healthy ranges. Discuss any dietary changes or concerns with your healthcare provider for personalized recommendations.

DEPARTMENT OF HAEMATOLOGY

Test Name	Value	Unit	Bio. Ref Interval
Absolute Eosinophil Count (AEC)	0.3	10^3/uL	0.02 - 0.50
ESR	17	mm/1st hour	0 - 12
Lymphocytes	35.2	%	20 - 40
Mean Corp Hb (MCH)	28.1	pg	27 - 32.0
Mentzer Index *<13: thalassemia trait more likely | ≥13: iron deficiency more likely*	17.21	Ratio	≥ 13
Absolute Monocyte Count	0.71	10^3/uL	0.20 - 1.00
Eosinophils	4.9	%	01 - 06
Absolute Neutrophil Count (ANC)	2.93	10^3/uL	2.00 - 7.00
Mean Corp Hb Conc (MCHC)	33.5	g/dL	31.5 - 34.5
RDW - CV	14.3	%	11.6 - 14.0
MPV	11.2	fL	7 - 9
Hematocrit (PCV)	41	%	40.0 - 50.0
Platelet Count(PLT)	155	10^3/µl	150 - 410
Absolute Basophil Count	0.02	10^3/uL	0.02 - 0.10
Absolute Lymphocyte Count (ALC)	2.15	10^3/uL	1.00 - 3.00
Monocytes	11.6	%	02 - 10
Basophils	0.4	%	00 - 02
Neutrophils	47.9	%	40 - 80
Haemoglobin (HB)	13.7	g/dL	13.0 - 17.0
RDW - SD	40.1	fL	39.0 - 46.0
Mean Corp Volume (MCV)	84	fL	83.0 - 101.0

Vitamin Profile

This section presents your **Vitamin and Micronutrient Profile**, evaluating critical nutritional markers essential for immune regulation, bone mineralisation, neurological integrity, and cellular energy metabolism. Vitamin D and Vitamin B12 deficiencies are among the most prevalent yet underdiagnosed contributors to fatigue, bone loss, immune suppression, and mood disturbance. Values shown in **red** indicate clinically significant deficiencies requiring targeted supplementation and dietary correction.

Clinical Implications

Your bone health is mildly affected, primarily indicated by the significantly elevated homocysteine level of 25.0 µmol/L, which is above the reference range of 3.7 - 13.9 µmol/L. This elevation poses cardiovascular and thrombotic risks and may suggest deficiencies in B vitamins such as B12, B6, and folate. Additionally, your vitamin D level is low at 26.69 ng/ml, falling below the optimal range of 30.0 - 100.0 ng/ml, which is crucial for calcium absorption and bone health. The serum calcium level of 9.5 mg/dl is optimal, indicating that calcium levels are currently adequate, but vitamin D insufficiency could hinder bone health over time.

Recommended Action

To address your elevated homocysteine, consider discussing supplementation of methylated B12 (methylcobalamin), B6 (pyridoxine), and folate with your physician, as these may help lower homocysteine levels. Recheck these levels in 8-12 weeks after starting supplementation. For your low vitamin D, aim for increased sun exposure and consider discussing vitamin D supplementation with your healthcare provider. Additionally, incorporate foods rich in B vitamins, such as leafy greens, legumes, and fortified cereals, into your diet. Regular weight-bearing exercises will also support bone health. Schedule a follow-up appointment to monitor your vitamin levels and overall bone health.

DEPARTMENT OF IMMUNOLOGY			
Test Name	**Value**	**Unit**	**Bio. Ref Interval**
VITAMIN B12	**491**	pg/ml	211 - 912
VITAMIN D (25 - OH VITAMIN D)	**26.69**	ng/ml	30 - 100

Thyroid Profile

This section presents your **Thyroid Function Profile**, evaluating the hormonal axis that governs your metabolic rate, energy levels, body temperature, and cardiac rhythm. TSH (Thyroid Stimulating Hormone), Free T3, and Free T4 are assessed to detect overt or subclinical hypothyroidism, hyperthyroidism, or autoimmune thyroid dysfunction. Values shown in **red** indicate hormonal imbalance that warrants medical review, as untreated thyroid dysfunction affects virtually every organ system in the body.

Clinical Implications

Your thyroid profile shows TSH, free T3, and free T4 within normal reference ranges, indicating a well-regulated thyroid axis and stable basal metabolic rate. There are no signs of hypothyroidism (low T3/T4, elevated TSH) or hyperthyroidism (elevated T3/T4, suppressed TSH). This reflects healthy pituitary-thyroid feedback and adequate iodine and selenium status.

Recommended Action

To maintain thyroid health, ensure adequate iodine intake through iodised salt, seafood, and dairy products. Avoid excessive consumption of raw goitrogenic foods such as raw cabbage, broccoli, and soy if thyroid status is borderline. Manage stress effectively, as chronic cortisol elevation can suppress thyroid function over time. Recheck TSH annually, especially if fatigue, weight changes, or hair loss occur.

DEPARTMENT OF IMMUNOLOGY			
Test Name	**Value**	**Unit**	**Bio. Ref Interval**
Thyroid Stimulating Hormone (TSH)-Ultrasensitive	**3.227**	µIU/ml	0.55 - 4.78
Tri-Iodothyronine (T3, Total)	**0.91**	ng/ml	0.60 - 1.81
Thyroxine (T4, Total)	**8.3**	ug/dl	3.2 - 12.6

HELIXLINE

Meal Plan - Week 1

Day	PRE BREAKFAST	BREAKFAST	SNACK 1	LUNCH	SNACK 2	DINNER
1	Methi seeds water + 1 tsp flaxseeds Methi seeds water	Moong dal cheela (2) + mint chutney — high protein Egg white omelet (4 whites) + sautéed mushrooms + spinach (no roti)	Small apple / guava / pomegranate (small bowl) Apple / Guava / Pomegranate (small bowl)	Quinoa (small) + moong dal + vegetables + salad (low GI) Grilled fish (150g) + moong dal soup + large mixed salad (no rice)	Roasted chana (small handful) Boiled egg whites (2-3)	Vegetable soup + dal (no roti) — low calorie, filling Grilled fish (120g) + vegetable soup + large salad (no roti)
2	Warm lemon water Lemon water	Besan cheela (2) + sautéed vegetables (high protein, lower GI) Boiled eggs (2) + moong dal cheela (1 piece) — high protein start	Buttermilk (low-fat, unsweetened) Green tea	1-2 multigrain roti + dal + palak sabzi + cucumber raita (low-fat) Grilled chicken breast (120g) + sautéed vegetables + dal soup (no roti)	Boiled chana chaat	Moong dal (thick) + sautéed vegetables (no grain — low carb) Chicken breast (100g) + dal + salad (no roti)
3	Green tea with cinnamon	Paneer bhurji (80g) + sautéed spinach + tomatoes (grain-free) Scrambled egg whites (4) + bell peppers + tomatoes (no grain)	Green tea + 10 almonds Cinnamon tea / turmeric water	Bajra roti (1) + dal + karela sabzi + salad (low GI grain) Egg curry (2 eggs - 1 yolk) + 1 bajra roti + large salad	Sprouts with lemon	Grilled paneer (50g) + 1 roti + large salad Fish tikka (120g) + raita + large green salad (no roti)
4	Ginger infused water 50-80ml black coffee (no sugar)	Tofu scramble (100g) + bell peppers + mushrooms (grain-free) Pan-grilled fish (60g) + 2 boiled eggs + mixed salad	Cinnamon tea / turmeric water Almonds (8-10)	Barley khichdi + buttermilk + salad (low GI) Steamed fish (150g) + stir-fried broccoli + cucumber raita (no grain)	Green tea Green tea + walnuts (5-6)	Dal palak + 1 multigrain roti + cucumber Steamed chicken breast (100g) + moong dal + cucumber salad
5	Tulsi infused water Ginger infused water	Soya chunks (50g dry weight) + stir-fried vegetables (high protein) Egg bhurji (3 eggs, 1 yolk) + sautéed vegetables (no roti — grain-free)	Roasted chana (small handful) Buttermilk (low-fat)	Rajma (large portion) + salad + cucumber raita (no roti — high protein) Clear chicken soup (large) + boiled egg (1) + steamed vegetables	Cinnamon tea	Tofu stir-fry + steamed vegetables (grain-free — anti-inflammatory) Baked turmeric-ginger fish (120g) + steamed broccoli (anti-inflammatory)
6	50-80ml black coffee (no sugar) Tulsi infused water	Green smoothie (spinach + cucumber + amla + 1 tbsp chia + protein) Green smoothie (spinach + cucumber + amla + 1 tbsp chia) — low calorie	Carrot sticks + cucumber Roasted chana (handful)	Paneer (80g) + moong dal + large vegetable salad (grain-free) Grilled fish + bajra roti (1) + moong dal + salad — moderate carb	Walnuts (5) + almonds (6) Roasted chana (small handful)	Masoor dal + cauliflower sabzi + raita (no roti) Egg white curry (3 whites) + cauliflower rice + salad (grain-free)
7	Ajwain (carom seeds) water Fennel seeds (saunf) water	Greek-style hung curd (100g) + seeds mix + 1 small fruit Poached eggs (2) + avocado (¼) + cucumber + hung curd (50g)	Coconut water Carrot sticks	Soya chunks curry + 1 ragi roti + salad — high protein Tuna salad bowl (80g) + boiled eggs (1) + cucumber + hung curd	Roasted peanuts (small handful) Buttermilk with roasted jeera	Oats khichdi (small) with vegetables + buttermilk Grilled fish + stir-fried vegetables + dal soup (no roti)

Disclaimer: This meal plan is a general guideline based on your health profile. Consulting a qualified nutritionist is recommended before following any meal plan. Individual nutritional needs may vary based on medical conditions, medications, and personal health goals.

HELIXLINE

Meal Plan - Week 2

Day	PRE BREAKFAST	BREAKFAST	SNACK 1	LUNCH	SNACK 2	DINNER
1	Ginger infused water 50-80ml black coffee (no sugar)	Tofu scramble (100g) + bell peppers + mushrooms (grain-free) Pan-grilled fish (60g) + 2 boiled eggs + mixed salad	Coconut water Carrot sticks	Barley khichdi + buttermilk + salad (low GI) Steamed fish (150g) + stir-fried broccoli + cucumber raita (no grain)	Roasted peanuts (small handful) Buttermilk with roasted jeera	Dal palak + 1 multigrain roti + cucumber Steamed chicken breast (100g) + moong dal + cucumber salad
2	Tulsi infused water Ginger infused water	Soya chunks (50g dry weight) + stir-fried vegetables (high protein) Egg bhurji (3 eggs, 1 yolk) + sautéed vegetables (no roti — grain-free)	Dark chocolate (1-2 squares, 70%+ cacao — dual cardio + insulin) Dark chocolate (1-2 squares, 70%+ cacao)	Rajma (large portion) + salad + cucumber raita (no roti — high protein) Clear chicken soup (large) + boiled egg (1) + steamed vegetables	Dark chocolate (2 squares, 70%+) + cinnamon tea Dark chocolate (2 squares, 70%+) + green tea	Tofu stir-fry + steamed vegetables (grain-free — anti-inflammatory) Baked turmeric-ginger fish (120g) + steamed broccoli (anti-inflammatory)
3	50-80ml black coffee (no sugar) Tulsi infused water	Green smoothie (spinach + cucumber + amla + 1 tbsp chia + protein) Green smoothie (spinach + cucumber + amla + 1 tbsp chia) — low calorie	Flaxseed water (1 tsp ground flax — omega-3 + fiber)	Paneer (80g) + moong dal + large vegetable salad (grain-free) Grilled fish + bajra roti (1) + moong dal + salad — moderate carb	Mixed seeds (flax + sunflower, 1 tbsp — omega-3, Vit E)	Masoor dal + cauliflower sabzi + raita (no roti) Egg white curry (3 whites) + cauliflower rice + salad (grain-free)
4	Ajwain (carom seeds) water Fennel seeds (saunf) water	Greek-style hung curd (100g) + seeds mix + 1 small fruit Poached eggs (2) + avocado (¼) + cucumber + hung curd (50g)	Roasted pumpkin seeds (1 tbsp — magnesium, zinc) Roasted pumpkin seeds (1 tbsp)	Soya chunks curry + 1 ragi roti + salad — high protein Tuna salad bowl (80g) + boiled eggs (1) + cucumber + hung curd	Walnut halves (4-5) + cinnamon tea (omega-3, insulin) Walnut halves (4-5) + cinnamon tea	Oats khichdi (small) with vegetables + buttermilk Grilled fish + stir-fried vegetables + dal soup (no roti)
5	Methi seeds water + 1 tsp flaxseeds Methi seeds water	Moong dal cheela (2) + mint chutney — high protein Egg white omelet (4 whites) + sautéed mushrooms + spinach (no roti)	Amla juice (small glass — Vit C, antioxidant) Amla juice (small glass)	Quinoa (small) + moong dal + vegetables + salad (low GI) Grilled fish (150g) + moong dal soup + large mixed salad (no rice)	Pomegranate seeds (small bowl — nitric oxide, antioxidant) Pomegranate seeds (small bowl)	Vegetable soup + dal (no roti) — low calorie, filling Grilled fish (120g) + vegetable soup + large salad (no roti)
6	Warm lemon water Lemon water	Besan cheela (2) + sautéed vegetables (high protein, lower GI) Boiled eggs (2) + moong dal cheela (1 piece) — high protein start	Brazil nuts (2-3) — selenium for thyroid support	1-2 multigrain roti + dal + palak sabzi + cucumber raita (low-fat) Grilled chicken breast (120g) + sautéed vegetables + dal soup (no roti)	Roasted chana (small handful) Seaweed broth / nori snack (iodine — thyroid support)	Moong dal (thick) + sautéed vegetables (no grain — low carb) Chicken breast (100g) + dal + salad (no roti)
7	Green tea with cinnamon	Paneer bhurji (80g) + sautéed spinach + tomatoes (grain-free) Scrambled egg whites (4) + bell peppers + tomatoes (no grain)	Pumpkin seeds (1 tbsp) + sunflower seeds — selenium, zinc Tuna in water (50g) + cucumber — selenium + iodine	Bajra roti (1) + dal + karela sabzi + salad (low GI grain) Egg curry (2 eggs - 1 yolk) + 1 bajra roti + large salad	Boiled chana chaat Turmeric milk (small cup, low-fat) — anti-inflammatory	Grilled paneer (50g) + 1 roti + large salad Fish tikka (120g) + raita + large green salad (no roti)

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“Thank You!”

This isn’t the end of a test. It’s the beginning of your healthiest chapter.

We’re honoured to be part of your journey, so you can live with precision, not guesswork.

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